

Best Practices for Data Presentation in Scientific Writing

Table of Contents

Introduction

General Principles for Data Presentation

Structure and Organization of Results Sections

Tables and Figures Best Practices

Reproducibility and Documentation Standards in Geosciences

Teaching and Learning Effective Data Writing

Introduction

Effective scientific writing in data sections requires clear presentation of purpose, process, and findings, with emphasis on reproducibility and pedagogical clarity. Students need specific practice writing data sections rather than general writing courses, as prior scientific writing experience is the only reliable predictor of success. (3 sources)

Scientific writing in data sections represents a critical component of research communication that extends far beyond simply reporting numerical results. A successful Results section must weave together three essential aspects: the purpose of the research, the process used to obtain findings, and the actual findings themselves ([Hood-DeGrenier, 2016](#)). This integration ensures that readers can understand not just what was discovered, but why it matters and how it was determined.

The importance of rigorous data presentation is particularly emphasized in geosciences and related fields, where scientific papers must demonstrate mathematical rigor, clarity, and pedagogical value in their writing and presentation ([Griffies et al., 2023](#)). Modern scientific communication demands that papers clearly articulate what was learned from the research while maintaining reproducibility and generality of results across all sections, from abstracts through conclusions and supplementary materials.

Training students in effective data writing requires targeted practice rather than general writing instruction. Research indicates that prior scientific writing experience serves as the only accurate predictor of student success in scientific writing, while general college-level writing courses, technical writing exposure, or years of study show no significant effect on students' ability to write research papers effectively ([Hood-DeGrenier, 2016](#)). This finding underscores the need for discipline-specific instruction where students analyze examples from actual scientific papers to learn general principles used by practicing scientists, moving beyond individual instructor preferences to understand broader scientific communication standards ([Spencer et al., 2018](#)).

Evidence

(Hood-DeGrenier, 2016)

[A Strategy for Teaching Undergraduates to Write Effective Scientific Results Sections](#)

"When discussing any research study, it is essential to understand the purpose, the process, and the findings. A successful Results section weaves together all three of these aspects"

"One study examining factors that influenced student success in scientific writing found that the only accurate predictor was prior scientific writing experience; neither the number of college-level writing courses, exposure to a technical writing course, nor the

number of years of study had any effect on the ability of students to write either a persuasive paper for a general audience or a research proposal (Jerde et al., 2004). Students need practice, and it is up to science instructors to require students to write in their classes and to seek the best ways to teach them to write effectively"

"Like Koen (5), I have found that students make the most progress in their writing when assignments are broken into small chunks. Rather than assigning a complete lab report, I ask them to prepare sections one at a time, providing instruction related to each section along the way."

(Griffies et al., 2023)

Aims and Scope of JAMES

"JAMES papers manifest scientific and mathematical rigor, clarity and pedagogy in writing and presentation, along with reproducibility and generality of results. JAMES papers clearly articulate what was learned as a result of performing the research. These attributes extend from the paper's summaries (both the scientific abstract and plain language abstract) to the methods, results, discussion, conclusions, appendices, and supplements, with each element serving to enhance the paper's longevity, impact, and readability."

(Spencer et al., 2018)

Preparing students for authentic research through open-inquiry laboratory courses

"our students write lab articles following the style of a typical research journal article with abstract, introduction, experimental materials and methods section, results, discussion, and conclusion. Students are expected to include correctly formatted references"

"We ask students to find examples of regular scientific papers that are relevant to the subject area to learn the general principles that scientists use to write papers rather than focusing on a specific teacher's preferences"

"we explain typical content of the various sections through regular in-class discussions of student papers that compare student writing to writing by practicing scientists. The instructor highlights what was done well, and how improvements can be made."

General Principles for Data Presentation

Effective data presentation requires organizing results into clear segments with purpose, approach, findings, and commentary while being selective about which data to include. Tables work best for precise data and multiple outcomes, while figures excel at showing trends and comparisons. (3 sources)

The foundation of effective data presentation rests on systematic organization and selective reporting of findings. Results sections should be structured into distinct segments where each presents the experimental purpose, approach, actual data through text and visuals, and meaningful commentary that highlights the most significant findings rather than simply repeating what appears in tables or figures ([Kallestinova, 2011](#)). This approach requires authors to distinguish between their complete dataset and the essential results needed for reader comprehension, often meaning that extensive experimental records collected over multiple trials should be condensed to focus only on findings critical for understanding ([Kallestinova, 2011](#)).

Two key principles guide effective data presentation: highlighting main findings and selecting appropriate formats based on data type ([Huston, 2020](#)). Tables prove most effective for presenting precise numerical data and multiple outcomes, with different populations organized in columns for comparison across variables listed in rows ([Huston, 2020](#)). Figures excel at displaying trends over time or comparative values, following standard conventions where independent variables appear on the horizontal axis and dependent variables on the vertical axis ([Huston, 2020](#)). Both formats benefit from concise titles that describe who, what, where, when, and how many elements of the study ([Huston, 2020](#)).

The primary function of results sections is presenting data in a systematic, concise, and visualized format while maintaining objectivity ([Kosycheva et al., 2022](#)). This systematic approach to data visualization and description enhances article impact, visibility in research databases, and significance within global scientific communication ([Kosycheva et al., 2022](#)).

Evidence

(Kallestinova, 2011)

How to Write Your First Research Paper

"Your Results should be organized into different segments or subsections where each one presents the purpose of the experiment, your experimental approach, data including text and visuals (tables, figures, schematics, algorithms, and formulas), and data commentary. For most journals, your data commentary will include a meaningful summary of the data presented in the visuals and an explanation of the most significant findings. This data presentation should not repeat the data in the visuals, but rather highlight the most important points"

"you should be selective in presenting data and choose only those experimental details that are essential for your reader to understand your findings. You might have conducted an experiment 20 times and collected numerous records, but this does not mean that you should present all those records in your paper. You need to distinguish your results from your data and be able to discard excessive experimental details that could distract and confuse the reader."

(Huston, 2020)

Effective presentation of data in tables and figures

"To present data effectively, you need to apply two key principles and certain best practices. The two key principles are to highlight the main findings and to choose the appropriate format based on the type of data. Best practices include conventions for creating titles; structuring tables and figures; and making abbreviations and footnotes accessible so they can be read by those who use electronic readers with text-to-speech technology"

"The best way to get the presentation of data in tables and figures just right is to use these to focus on the main findings, including both the results of the methodology (more on this below) and the main outcome measures of the study"

"Tables are most effective for presenting precise data and multiple outcomes. Figures are most effective for presenting trends over time or comparative values. When constructing a table, different populations are identified in columns and then compared according to variables that are identified in rows. This structure enables comparisons between the different study populations. When constructing figures, the independent variable (such as time) is on the horizontal or x-axis and the dependent or outcome variable is on the vertical or y-axis. Good titles for both tables and figures give a concise description of who, what, where, when and how many."

(Kosycheva et al., 2022)

Design Strategies in Research Articles Results Section

"The function of the Research Article Results section is to present the data obtained during the research in a systematic, concise, and visualized format. Commenting and discussing the information presented impedes objective perception and may confuse the reader"

"Based on the analysis of models for structuring the Research Article Results section from different fields, an efficient pattern of constructing a section is proposed and illustrated using the example from journals on food processing and health-saving technologies. Recommendations for data visualization and their relevant description in the body of the section are given"

"The outcomes of this editorial and proposed guidelines for constructing the Results section, are intended to help authors efficiently present the data collected during the study (logically, concisely, transparently, visualized and objectively). Such an approach not only makes it possible to increase the impact of the article on the readership, but also contributes to its visibility in research databases, enhances its significance in the context of global scientific communication."

Structure and Organization of Results Sections

Results sections should be organized into clear subsections following a logical chronological order, with each segment presenting experimental purpose, approach, data, and commentary. Methods sections support this by providing structured descriptions that enable reproducibility and validity assessment. (4 sources)

Effective organization of results sections requires a systematic approach that divides content into distinct subsections, each containing four essential components: the purpose of the experiment, the experimental approach, the actual data (including text and visuals such as tables, figures, schematics, algorithms, and formulas), and meaningful data commentary (Kallestinova, 2011). This structured approach ensures that readers can follow the logical progression from research questions to findings while understanding the methodology behind each result.

The organizational framework should follow a chronological structure that moves from theoretical elements to numerical aspects to analysis and post-processing (Zelst et al., 2021). This logical sequence helps readers

understand both what was done to answer the research question and how it was accomplished, supporting the overall validity of the research ([Kallet, 2004](#)). When presenting large amounts of detail, information should be organized into subsections according to topic, with material in each section arranged from most to least important ([Kallet, 2004](#)).

A well-structured results section begins with a brief outline paragraph that describes the study design and main steps taken to answer the scientific question posed in the introduction ([Zelst et al., 2021](#)). This organizational approach should address the fundamental questions of who performed the experiment, what was done, when and where it was undertaken, how it was executed and analyzed, and why specific procedures were chosen ([Zelst et al., 2021](#)). The ultimate goal is presenting data in a systematic, concise, and visualized format that maintains objectivity and enhances the article's impact and visibility in research databases ([Kosycheva et al., 2022](#)).

Evidence

(Kallestinova, 2011)

[How to Write Your First Research Paper](#)

"Your Results should be organized into different segments or subsections where each one presents the purpose of the experiment, your experimental approach, data including text and visuals (tables, figures, schematics, algorithms, and formulas), and data commentary. For most journals, your data commentary will include a meaningful summary of the data presented in the visuals and an explanation of the most significant findings. This data presentation should not repeat the data in the visuals, but rather highlight the most important points"

"you should be selective in presenting data and choose only those experimental details that are essential for your reader to understand your findings. You might have conducted an experiment 20 times and collected numerous records, but this does not mean that you should present all those records in your paper. You need to distinguish your results from your data and be able to discard excessive experimental details that could distract and confuse the reader."

(Zelst et al., 2021)

[101 Geodynamic modelling: How to design, carry out, and interpret numerical studies](#)

"The methods section is considered one of the most important parts of any scientific manuscript (Kallet, 2004). A good methods section allows other scientists to verify results and conclusions, understand whether the design of the experiment is relevant for the scientific question (validity), and to build on the work presented (reproducibility and replicability) by assessing alternative methods that might produce differing results. Thus, the major goals of the methods section are to verify the experiment layout and allow others to reproduce the results"

"First, the methods should be plain and simple, objective, and logically described and should be thought of as a report of what was done in the study. Unstructured and incomplete methods can make the manuscript cumbersome to read or even lead the reader to question the validity of the research"

"The 'who, what, when, where, how, and why' order proposed by Annesley (2010) breaks the methods section down in the following questions: who performed the experiment? What was done to answer the research question? When and where was the experiment undertaken? How was the experiment done, and how were the results analysed? Why were specific procedures chosen?"

"The methods should start with a brief outline (one paragraph) describing the study design and the main steps taken to answer the scientific question posed in the introduction. The outline should be logical and go from theoretical elements, to numerical aspects, to analysis and post-processing."

(Kallet, 2004)

[How to Write the Methods Section of a Research Paper](#)

"The methods section of a research paper provides the information by which a study's validity is judged. Therefore, it requires a clear and precise description of how an experiment was done, and the rationale for why specific experimental procedures were chosen. The methods section should describe what was done to answer the research question, describe how it was done, justify the experimental design, and explain how the results were analyzed. Scientific writing is direct and orderly. Therefore, the methods section structure should: describe the materials used in the study, explain how the materials were prepared for the study, describe the research protocol, explain how measurements were made and what calculations were performed, and state which statistical tests were done to analyze the data. Once all elements of the methods section are written, subsequent drafts should focus on how

to present those elements as clearly and logically as possible. The description of preparations, measurements, and the protocol should be organized chronologically. For clarity, when a large amount of detail must be presented, information should be presented in sub-sections according to topic. Material in each section should be organized by topic from most to least important."

(Kosycheva et al., 2022)

Design Strategies in Research Articles Results Section

"The function of the Research Article Results section is to present the data obtained during the research in a systematic, concise, and visualized format. Commenting and discussing the information presented impedes objective perception and may confuse the reader"

"Based on the analysis of models for structuring the Research Article Results section from different fields, an efficient pattern of constructing a section is proposed and illustrated using the example from journals on food processing and health-saving technologies. Recommendations for data visualization and their relevant description in the body of the section are given"

"The outcomes of this editorial and proposed guidelines for constructing the Results section, are intended to help authors efficiently present the data collected during the study (logically, concisely, transparently, visualized and objectively). Such an approach not only makes it possible to increase the impact of the article on the readership, but also contributes to its visibility in research databases, enhances its significance in the context of global scientific communication."

Tables and Figures Best Practices

Effective tables and figures focus on main findings rather than comprehensive data, with tables best for precise numerical data organized by populations in columns and variables in rows, while figures excel at showing trends with independent variables on the x-axis. Both require concise titles describing who, what, where, when, and how many. (1 source)

Creating effective tables and figures requires applying two fundamental principles: highlighting main findings and choosing the appropriate format based on data type (Huston, 2020). The most critical aspect is using these visual elements to focus on main findings, including both methodological results and primary outcome measures of the study, rather than attempting to present all collected data (Huston, 2020).

The choice between tables and figures should be guided by the nature of the data being presented. Tables prove most effective for presenting precise numerical data and multiple outcomes, with the optimal structure placing different study populations in columns to enable direct comparison across variables listed in rows (Huston, 2020). This columnar organization facilitates systematic comparison between different study populations across multiple measured parameters.

Figures excel at displaying trends over time or comparative values, following established scientific conventions where the independent variable appears on the horizontal x-axis and the dependent or outcome variable on the vertical y-axis (Huston, 2020). This standard arrangement helps readers immediately understand the relationship between variables being examined.

Both tables and figures benefit from well-crafted titles that provide concise descriptions covering the essential elements: who was studied, what was measured, where the study took place, when it was conducted, and how many subjects were included (Huston, 2020). Additional best practices include making abbreviations and footnotes accessible for electronic readers with text-to-speech technology, ensuring broader accessibility of the research findings (Huston, 2020).

Evidence

(Huston, 2020)

Effective presentation of data in tables and figures

"To present data effectively, you need to apply two key principles and certain best practices. The two key principles are to highlight the main findings and to choose the appropriate format based on the type of data. Best practices include conventions for creating titles; structuring tables and figures; and making abbreviations and footnotes accessible so they can be read by those who use electronic readers with text-to-speech technology"

"The best way to get the presentation of data in tables and figures just right is to use these to focus on the main findings, including

both the results of the methodology (more on this below) and the main outcome measures of the study"

"Tables are most effective for presenting precise data and multiple outcomes. Figures are most effective for presenting trends over time or comparative values. When constructing a table, different populations are identified in columns and then compared according to variables that are identified in rows. This structure enables comparisons between the different study populations. When constructing figures, the independent variable (such as time) is on the horizontal or x-axis and the dependent or outcome variable is on the vertical or y-axis. Good titles for both tables and figures give a concise description of who, what, where, when and how many."

Reproducibility and Documentation Standards in Geosciences

The Geoscience Paper of the Future initiative established 20 specific best practices covering data accessibility, software documentation, and method transparency to ensure computational research can be fully reproduced. These standards address the challenge that traditional publications cannot capture the full complexity of modern digital geoscience research. (2 sources)

The geoscience community has developed comprehensive standards to address reproducibility challenges in an era of digital data and computational methods. The Geoscience Paper of the Future (GPF) initiative established best practices to ensure research products including data, software, and methods are openly accessible, citable, and well documented (Gil et al., 2016). This approach recognizes that computational research cannot be fully captured in traditional publications alone.

The GPF framework includes 20 specific recommendations organized across seven key areas (Yu et al., 2016) (Gil et al., 2016):

- **Data accessibility** - Making datasets openly available and discoverable
- **Data documentation** - Providing comprehensive metadata and descriptions
- **Software accessibility** - Ensuring computational tools are available to other researchers
- **Software documentation** - Including clear instructions and code comments
- **Provenance documentation** - Recording the complete workflow from data to results
- **Method documentation** - Detailing analytical procedures and decision points
- **Author identification** - Using persistent identifiers for proper attribution

These standards enable scientists to manage research products as valuable scientific assets while empowering broader access by other scientists, students, decision makers, and the public (Gil et al., 2016). The ultimate goal is accelerating scientific discovery by improving the ability of others to build upon published work through transparent documentation and dissemination practices.

Paper	Focus Area	Proposed Best Practices	Scope Of Reproducibility	Target Audience For Practices	Motivation For Open Science	Approach To Documentation
Toward the Geoscience Paper of the Future: Best practices for documenting and sharing research from data to software to provenance	Documenting and sharing geoscience research products	Openly accessible, citable, and well-documented data, software, and methods.	data, software, methods, provenance	GPF authors, scientists, students, decision-makers, public	Accelerating discovery, broader access, and transparency .	Document and cite all research products: data, software, provenance.

Paper	Focus Area	Proposed Best Practices	Scope Of Reproducibility	Target Audience For Practices	Motivation For Open Science	Approach To Documentation
Open science in practice: Learning integrated modeling of coupled surface-subsurface flow processes from scratch	Integrated modeling of coupled surface-subsurface flow	Documenting data, software, workflow in research papers	Documentation of data, software, and computational workflow	Earth science and engineering users, new users	Promotes reusability, collaboration, utility of software and papers.	Documentation of data, software, and computational workflow

Evidence

(Gil et al., 2016)

[Toward the Geoscience Paper of the Future: Best practices for documenting and sharing research from data to software to provenance](#)

"This article proposes best practices for GPF authors to make data, software, and methods openly accessible, citable, and well documented. The publication of digital objects empowers scientists to manage their research products as valuable scientific assets in an open and transparent way that enables broader access by other scientists, students, decision makers, and the public. Improving documentation and dissemination of research will accelerate the pace of scientific discovery by improving the ability of others to build upon published work."

(Yu et al., 2016)

[Open science in practice: Learning integrated modeling of coupled surface-subsurface flow processes from scratch](#)

"The GPF activity proposed best practices to make sure that geoscience papers of the future would capture the core concept of reproducibility (Gil et al., 2016). These best practices consist of 20 recommendations focusing on data accessibility, data documentation, software accessibility, software documentation, provenance documentation, method documentation, and author identification."

Teaching and Learning Effective Data Writing

Students learn data writing best through practice with actual journal articles rather than general writing courses, with assignments broken into small sections and regular comparison between student work and professional scientific writing. Science instructors must provide targeted instruction since prior scientific writing experience is the only predictor of success. (2 sources)

Effective instruction in data writing requires a structured approach that immerses students in authentic scientific communication practices. Students should write lab articles following the style of typical research journal articles, including abstracts, introductions, experimental materials and methods sections, results, discussions, and conclusions with correctly formatted references ([Spencer et al., 2018](#)). Rather than focusing on individual instructor preferences, students should examine examples of actual scientific papers relevant to their subject area to learn the general principles that practicing scientists use in their writing ([Spencer et al., 2018](#)).

The most effective teaching approach involves regular in-class discussions that compare student writing directly to professional scientific writing. Instructors should highlight what students have done well while identifying specific areas for improvement through systematic comparison with published research ([Spencer](#)

[et al., 2018](#)). This method helps students understand that successful data writing integrates three essential aspects: the purpose of the research, the process used to obtain findings, and the actual findings themselves ([Hood-DeGrenier, 2016](#)).

Breaking assignments into manageable components proves more effective than assigning complete lab reports. Students make the most progress when they prepare sections one at a time, with targeted instruction provided for each section along the way ([Hood-DeGrenier, 2016](#)). This approach acknowledges that students need extensive practice in scientific writing, making it essential for science instructors to require writing in their classes and seek the most effective teaching methods for their students ([Hood-DeGrenier, 2016](#)).

Evidence

(Spencer et al., 2018)

[Preparing students for authentic research through open-inquiry laboratory courses](#)

"our students write lab articles following the style of a typical research journal article with abstract, introduction, experimental materials and methods section, results, discussion, and conclusion. Students are expected to include correctly formatted references"

"We ask students to find examples of regular scientific papers that are relevant to the subject area to learn the general principles that scientists use to write papers rather than focusing on a specific teacher's preferences"

"we explain typical content of the various sections through regular in-class discussions of student papers that compare student writing to writing by practicing scientists. The instructor highlights what was done well, and how improvements can be made."

(Hood-DeGrenier, 2016)

[A Strategy for Teaching Undergraduates to Write Effective Scientific Results Sections](#)

"When discussing any research study, it is essential to understand the purpose, the process, and the findings. A successful Results section weaves together all three of these aspects"

"One study examining factors that influenced student success in scientific writing found that the only accurate predictor was prior scientific writing experience; neither the number of college-level writing courses, exposure to a technical writing course, nor the number of years of study had any effect on the ability of students to write either a persuasive paper for a general audience or a research proposal (Jerde et al., 2004). Students need practice, and it is up to science instructors to require students to write in their classes and to seek the best ways to teach them to write effectively"

"Like Koen (5), I have found that students make the most progress in their writing when assignments are broken into small chunks. Rather than assigning a complete lab report, I ask them to prepare sections one at a time, providing instruction related to each section along the way."